



HUMAN- MACHINE *Communication*

Rethinking Communication,
Technology, and Ourselves

Edited by **Andrea L. Guzman**

From virtual assistants to social robots, people are increasingly interacting with intelligent and highly communicative technologies throughout their daily lives. This shift from communicating with people to communicating with people and machines challenges how scholars have theorized and studied communication. *Human-Machine Communication: Rethinking Communication, Technology, and Ourselves* addresses this transition in how people communicate and who, or what, they communicate with and the implications of this evolution for communication research. Geared toward scholars interested in people's interactions with technology, this book serves as an introduction to human-machine communication (HMC) as a specific area of study within communication (encompassing human-computer interaction, human-robot interaction, and human-agent interaction) and to the research possibilities of HMC. This collection includes papers presented as part of a scholarly conference on HMC, along with invited works from noted researchers. Topics include defining HMC, theoretical approaches to HMC, applications of HMC, and the larger implications of HMC for self and society. The research presented here focuses on people's interactions with multiple technologies (artificial intelligence, algorithms, and robots) used within different contexts (home, workplace, education, journalism, and healthcare) from a variety of epistemological and methodological approaches (empirical, rhetorical, and critical/cultural). Overall, *Human-Machine Communication* provides readers with an understanding of HMC in a way that supports and promotes further scholarly inquiry in a growing area of communication research.

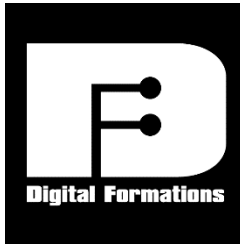
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Human-Machine Communication



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General Editor

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This book is dedicated to all of the selfless academic mentors I have been blessed to have in my life, from grade school through today. I will continue to do my best to pay forward your guidance, encouragement, and patience.



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Preface

This volume demarcates a new area of research within the larger study of communication—Human-Machine Communication (HMC). As I explain in greater detail in the introduction, the concept of human-machine communication is not new (it is more than a half-century old); however, the conceptualization of people’s interactions with various machines (artificial intelligence, robots, algorithms, industrial technologies, etc.) as *communication* by *communication scholars* with a collective focus on understanding the *creation of meaning between human and machine* through the *process of message exchange between human and machine*, the *building of relationships among people and technology*, and the resulting *individual, societal, and cultural implications* is novel.

This book is the product of the first HMC post-conference for communication scholars, “Communicating with Machines: The Rising Power of Digital Interlocutors in Our Lives,” that convened as part of the International Communication Association’s (ICA) conference in Fukuoka, Japan in June 2016. I spearheaded the conference along with Steve Jones, University of Illinois at Chicago (UIC); David J. Gunkel, Northern Illinois University (NIU); Autumn P. Edwards, Western Michigan University; Chad Edwards, Western Michigan University; and Patric R. Spence, University of Kentucky. The conference was sponsored by the Department of Communication at UIC; the Department of Communication at NIU; the College of Communication & Information, School of Information Science at the University of Kentucky; School of Communication, Western Michigan University, and the Communication & Social Robotics Labs. The inaugural all-day event included paper and poster presentations, networking, and formal and informal discussions around building HMC research within the discipline. Steve Jones generously offered for the annual Steve Jones Internet Research Lecture, sponsored by ICA and the Carl Couch Center for Social and Internet Research, to serve

as the keynote presentation. Leopoldina Fortunati delivered the keynote, “Feminism, Labor and the Mechanization of Everyday Life.” Most of the chapters within this volume are developed from papers or topics presented by scholars at this event or submitted to it. I invited Matthew Lombard and Charles Ess to contribute chapters in their areas of expertise because of their important theoretical and philosophical scholarship in this area. The goal of this volume is not only to capture the questions raised and ideas discussed during the post-conference, and, thus, serve as a historical text, but also to showcase HMC research as it is emerging within communication. Still, what is included here is only a portion of the 22 papers and posters presented during the post-conference and is an even smaller representation of HMC research throughout communication that is continuing to grow.

The history of this book and the establishment of this area of research begins well before Fukuoka, Japan in 2016. Many of the people who are part of what is now called HMC already had been studying and theorizing aspects of people’s communication with technology, including social agents and robots, for years prior to the post-conference. My own interest in this area began with Apple’s release of Siri with the iPhone 4s in late 2011. Siri captivated me as a consumer and as a doctoral student in UIC’s communication department. Thinking about my own interactions with Siri, observing media portrayals regarding Siri, and listening to people’s reactions to Siri sparked my research interest in people’s conceptualizations of and relationships to AI technologies as communicators. It was during the research for my dissertation, *Imagining the Voice in the Machine: The Ontology of Digital Social Agents*, that I became aware of crucial gaps in communication scholarship and the need for increased opportunities for dialogue among scholars regarding the study of communication with different forms of technology. That communication scholars needed to pay greater attention to people’s interactions with machines was something that Steve Jones, my PhD dissertation advisor at UIC, had been talking about for some time, as had been David Gunkel, my MA thesis advisor at NIU.

From my perspective, the most pressing matter surrounding the study of people’s communication with devices and applications and the easiest to start rectifying was to get scholars doing this work into the same space to begin that scholarly dialogue. In late 2014, I approached Steve Jones about proposing a Blue Sky Workshop regarding human-machine communication as part of ICA’s annual conference. The goal of Blue Sky workshops is to give scholars a space to discuss shared issues or areas of research. David Gunkel was invited to serve as our third chair for the workshop, and I drafted the proposal. Several months later (and only a few hours after I had submitted my dissertation to my committee), Steve, David, and I were leading the

workshop, “Beyond Human: Developing Human-Machine Communication Research,” in May 2015, in San Juan, Puerto Rico. It was a beautiful afternoon in the Caribbean, but the workshop was full, with several dozen people in attendance. Many of the attendees had similar experiences as Steve, David, and myself: they had been working in areas related to HMC and were looking to connect with a larger body of scholars with similar interests.

The shared sentiment at the workshop was to move forward with a more formal event, specifically a pre or post-conference at ICA the following year. Now in my first year as an assistant professor at NIU, I initiated the planning along with David and Steve, and we agreed to reach out to several workshop participants who expressed interest in assisting with the preconference, specifically Autumn P. Edwards and Chad Edwards from Western Michigan University. Autumn and Chad had founded the Communication and Social Robotics Labs along with Patric R. Spence, who was then at the University of Kentucky. These combined efforts were behind the post-conference in Japan.

Efforts to formalize HMC have continued to move forward as more scholars have become interested and involved in the study of people’s communication with technology. Anecdotally, one of the factors driving this interest in HMC is that technology now talks back—literally—and this talkative technology is making its way into almost every aspect of people’s lives. Technologies such as digital agents and social robots also are beginning to make way for HMC to be accepted as an area of study within communication. One of the reasons why this area of research is only now taking shape is that scholars initially faced resistance to the idea of machines as *communicators*. The proliferation of talking and moving technology, particularly devices and applications supported by AI, has meant that even scholars outside of HMC have experienced a communicative technology, helping them to better understand our arguments and the need for HMC research.

Given the success of the first ICA event, we convened a second ICA preconference, “Communicating with Machines: Interventions with Digital Agents,” in May 2017 in San Diego, California. Autumn P. Edwards and Chad Edwards served as lead organizers with Patric R. Spence, Steve Jones, David J. Gunkel, and myself once again as co-chairs. Austin Lee, who was then at Northern Kentucky University, and Jake Liang, of Chapman University also joined as co-chairs, as did Seth C. Lewis, of the University of Oregon. Our list of sponsors also grew: Department of Communication, NIU; College of Informatics, Northern Kentucky University; Department of Communication, UIC; College of Communication & Information, School of Information Science, University of Kentucky; Shirley Papé Chair in Emerging Media, School of Journalism and Communication, University of Oregon; Communication

and Social Robotics Labs, School of Communication, Western Michigan University. With the help of Matthew Lombard and his resources at Temple University, I also have established an e-mail list to help better connect HMC scholars, and we also have convened informal meetings of HMC scholars at conferences beyond ICA, including the Association of Internet Researchers. As this book is going to print, we now are preparing for our third ICA preconference “Communicating with Machines: Theory and Practice” with Patric R. Spence at the helm, to be held in Prague in May 2018. The number of submissions has almost tripled from Fukuoka.

While our community has gained so much, I would like to take a moment to also acknowledge one of its losses: In 2017, Jake Liang who had worked with Austin Lee in studying people’s interactions with robots passed away. The chapter authored by Austin and Jake in this volume was written before Jake’s passing. We are all saddened at the loss of Jake, a wonderful person and an excellent scholar. The chapter written by Austin and Jake brings together so much of their work and provides the opportunity to share Jake’s passion with a new audience.

The purpose of recounting this history is to highlight how quickly interest in HMC has developed and to underscore the importance of cultivating a community of scholars that provides the space not only to present one’s work but also to engage in discussion regarding larger issues and questions. One of the goals we have set as organizers of HMC events and facilitators of this group has been to encourage dialogue across the topical, epistemological, and methodological divides within the study of communication. This is not a group of qualitative or quantitative scholars. It is a community of qualitative and quantitative scholars, rhetoricians, philosophers (and the list goes on) who are seeking answers to some of the most pressing questions facing individuals, society, and the study of communication itself. (After all, if we have long defined communication as a human-only process, the shift toward theorizing it as a human-machine process challenges the very nature of communication and communication research).

My goal in developing this edited volume has been to provide a collection of scholarship that is representative of the work of individual HMC scholars presented at the first post-conference, but, at the same time, attempts to capture the breadth of HMC research and approaches to it. To that end, the scholarship included in this volume focuses on people’s communication with different technologies (e.g., AI, various types of robots, algorithms), proceeds from multiple epistemological and theoretical starting points (e.g., social-scientific, critical, cultural), and uses different methods (e.g., survey, experiment, rhetorical analysis, philosophical inquiry). At the same time, I

realize that many audience members may be new to HMC and are reading this book to inform their own research, to incorporate its material into classes, or to satisfy a general curiosity in this area. I have charged myself and the chapter authors to be clear in how individual research relates to HMC overall as well as to pay attention to issues of theory and method.

The result is that this book can be read several ways. The first is that readers can focus on a particular technology, theory, or method that is most relevant to their own interests. Such a reading would focus on a single or a few chapters. The second way to read this book is in its entirety, or near entirety. It is this second approach that I suggest (and, no, I am not suggesting this approach because I am the editor). I advocate for a more holistic reading because it best recreates, as much as is feasible in print form, the nature of the HMC conference at which this research was presented. To read only a chapter or two that is most relevant to one's research is efficient, no doubt, but keeps firmly in place those divisions that in many ways have hampered the study of HMC in the first place. Reading this book in its entirety will help individuals, particularly people who are new to this area, to (1) understand the individual work taking place, (2) identify areas of convergence and divergence within the scholarship, and (3) gain a greater appreciation of the overarching issues and questions underpinning this research as a whole.

The one aspect of the HMC community that I, unfortunately, cannot adequately capture within printed text is its professional congeniality and shared comradery. Yes, as scholars we do have our philosophical, theoretical, and methodological differences, and, yes, these differences matter. However, what we as organizers of this community and many of its participants share is an understanding that collectively we are working toward something much bigger than our individual research agendas. It is through our own research that we have realized that what we know of communication is fundamentally changing as machines enter into our communicative spaces. We are beginning to see the very early stages of how communicative technologies are transforming our understanding of not only what machines are but also who we are. For these reasons, we understand that the study of HMC is clearly bigger than any one of us or any single approach to research. The final purpose of this book, then, is to invite its readers to join us in tackling these questions surrounding people's communication with machines and to bring new questions, new theories, and new approaches to this endeavor.

Andrea L. Guzman
DeKalb, Illinois
February 2018



Acknowledgments

The development of Human-Machine Communication as an area of study has been a collective effort. My gratitude goes foremost to Steve Jones and David J. Gunkel who have been key collaborators in formally establishing and growing the study of HMC. Steve and David each have served as my advisor during different points in my academic career and still provide me with invaluable mentorship for which I am grateful. I also am glad that Autumn P. Edwards, Chad Edwards, and Patric R. Spence have joined us in this endeavor. They and the Communication and Social Robotics Labs they direct have been invaluable to building the HMC community. Part of this growth also has included contributions from Austin Lee and Jake Liang. I am thankful to all of you for providing your time and talent to facilitating the formation of HMC and leading HMC events, but, most of all, I am humbled by your encouragement and ongoing support.

Several members of the HMC community also deserve special acknowledgement. During one of our early conversations, Seth C. Lewis asked me: “What is Human-Machine Communication, anyway?” It is a fundamental question, if not *the* fundamental question, of HMC that I take up in this book’s introduction. My answer to that question is better today because of my collaboration with Seth on related projects and our ongoing dialogue—particularly more of his excellent questions—around the theoretical dimensions of HMC. Gina Neff also has been an excellent mentor to me: It was Gina’s initial excitement at the workshop in Puerto Rico that gave me the confidence as a junior scholar to continue pursuing HMC. Matthew Lombard has generously given of his time and resources to help us establish a formal e-mail list that has helped us to grow and sustain this group. Eleanor Sandry has had one of the hardest tasks: helping me keep my type-A personality in check with her wit and her wisdom.

My work in HMC and toward this book has spanned my time at UIC as a doctoral student and now at NIU as an assistant professor. Thanks goes to my professors and peers at UIC who guided and supported me as I started

my research in this area. The professional, financial, and personal support of Northern Illinois University and my colleagues in the department of communication played an enormous role in making HMC events and this book a reality. This book also was financially supported by a NIU Research and Artistry Facilitation Grant.

As an edited volume, this book would not be possible without its contributors and their hard work in writing thought-provoking submissions. Putting together a book featuring more than a dozen authors is not an easy task, but it was made infinitely more manageable with copyediting help from Kandice Jenkins and indexing assistance from Parisa Farhadi. Kathryn Harrison at Peter Lang also deserves a word of thanks for patiently working with this first-time book editor.

The contributions of my family and friends are too many to list here, and there are not enough words to thank you all for your ongoing support. My husband is still the best person around. And, Thumper, you deserve an extra treat.



Introduction: “What Is Human-Machine Communication, Anyway?”

ANDREA L. GUZMAN

The question that serves as this chapter’s title is the guiding theme for this edited volume. It is one that I and other human-machine communication scholars have been asked numerous times as we have attempted to formalize this area of study and its central concept. Most people grasp that, as its name implies, human-machine communication, or HMC, has something to do with people’s interactions with technology. In actuality, human-machine communication is both a concept and an area of research within communication. It is the *creation of meaning among humans and machines* and the study of this meaning-making and related aspects. But in a discipline where communication has long been conceptualized as a human process through machines, exactly what human-machine communication is, where it fits into the study of communication, and what it can potentially bring to communication are not readily clear. HMC as an area of study within communication also is still taking shape. Its primary catalyst has been scholars from throughout the discipline coming together to discuss their research, to address shared questions, and to help build a scholarly community. Because HMC is taking shape from the ground up, questions related to the nature of HMC are just as critical for existing HMC scholars as they are for scholars and students new to this area.

This chapter begins to provide some of these answers. I say, begin, because this book is only one step, but an important one, toward codifying this area of study. The conceptualization of HMC presented is not mine alone. To address questions regarding the nature of HMC, I have studied more than seven decades of research regarding the idea of communication between people and technology both inside and outside the discipline. I have listened to how other HMC scholars conceptualize what it is that they do and have

paid careful attention to the ways in which this volume's authors explained their contributions. In staking out what HMC is within the coming pages, I provide a glimpse into what has influenced the shape of HMC so that other scholars can build upon these ideas. Toward this end, I begin by orienting the reader to HMC in relation to communication generally. Next, I discuss the origins of HMC outside the discipline and within it, before explaining why HMC as a new area of study is needed. Finally, I more thoroughly explain and define HMC while introducing this volume's chapters that exemplify HMC research.

Questions of Communication

For many people, communication is understood as a process taking place between themselves and someone else. To communicate means posting to social media followers, texting a loved one, talking with a friend, giving a presentation to colleagues, or reading the latest news written by a journalist. Communication also is, and has been, defined as a human-only process by those who study it. As the study of communication was coalescing in the mid-twentieth century, influential scholars purposefully drew its boundaries around human interaction (e.g., Schramm, 1973). In textbooks, communication is presented within a primarily, if not exclusively, human context, with models, theories, and examples focused on people's interactions. A visual of the transactional model of communication depicting messages flowing between two human-like figures greets visitors to the "What is Communication?" section of the National Communication Association's website.¹ Representations of mass communication historically have replaced the face-to-face image of interpersonal contexts with a depiction of a single source—a media organization composed of people—transmitting information to a *human* audience. In this default conceptualization of communication, technology is the medium or channel. It is the mobile phone people use to text that loved one or a specific app that delivers the day's news.

Questioning the Role of Technology

As I explain in further detail later, many emerging technologies no longer readily fit into this role of medium only. They are designed to function as things with which people interact. The mobile phone is not only a means for contacting friends but also a talkative assistant that helps people carry out tasks. That news app doesn't deliver just any story; it "learns" people's preferences, providing individualized content. Some of the news stories coming

through that app now are being created by computer programs instead of reporters. And the robots people once encountered only in science fiction are moving into their homes.

It is these exchanges between people and technology that are examples of human-machine communication and are the focus of this area of research. *In human-machine communication, technology is conceptualized as more than a channel or medium: it enters into the role of a communicator.* In contrast to models of human communication, HMC models have messages flowing between and among humans and machines. The “image” of communication shifts from humans facing each other to a human facing a machine. Described this way, HMC seems relatively straightforward, the swapping of a machine for a human within an interaction. Indeed, substituting a technology for a person in experiments that test a communication or social theory has been one of the important ways scholars have studied people’s interactions with computers (e.g., Reeves & Nass, 1998).

Questioning Meaning-Making With Technology

However, communication is conceptualized as more than the study of roles in a process and the transmission of information (Carey, 1989). Central to the development of communication theory is a focus on the meaning of messages and their effects (Rogers, 1997) and, as Schramm (1973) explains, how this meaning enters into people’s relationships with one another:

When we study communication, therefore, we study people—relating to each other and to their groups, organizations, and societies, influencing each other, being influenced, informing and being informed, reaching and being taught, entertaining and being entertained. To understand human communication, we must understand how people relate to one another. (pp. 3–4)

Communication also is the means through which people learn about their world (Blumer, 1969), form an understanding of Self and Other (Mead, 1967) and contribute to the shape of society (Cooley, 1897/2004). Communication research, then, is about who we are, who we are to one another, and the very reality that we are creating (Carey, 1989).

From this perspective, the view of human-machine communication switches from that of a process between a person and technology to the *creation of meaning between human and machine*. The questions surrounding HMC now seem much more complex. And, they are. How do messages from robots affect us? What sorts of relationships emerge when a technology becomes a communicator? How do people understand themselves as the result of their interactions with a virtual agent? What type of society is being

constructed through people's ongoing communication with both humans and machines? What is communication now that it extends to our interactions with machines? These are some of the many complex questions that human-machine communication scholars seek to answer.

How Have We Theorized Communicating With Machines?

The study of people's interactions with technology has an extensive history, and the current direction of HMC and its conceptualization have been influenced by research inside communication and scholarship outside of it. This section briefly reviews those influences by focusing on key points of theoretical intersection and divergence.² Overall, engineers and computer scientists have used the idea of communication to help guide technology design and development of technological interfaces with a goal of facilitating use. Meanwhile, the study of a specific medium or research regarding people's direct interactions with media have had a place within the study of communication, but this research has received considerably less attention because it was outside communication's prevailing anthropocentric paradigm.

Engineering, Computers, and Communication

While many communication scholars are cognizant of the theoretical intersections between communication and fields within the social sciences and the humanities, they may be less aware of the importance of the idea and theories of communication to researchers within engineering and technology. That the discipline's foundational model (Shannon & Weaver, 1949) had its roots in engineering is an often overlooked connection between communication and technological fields, a point I will return to later. Since the mid-twentieth century, engineers, computer scientists and other researchers have approached people's use of computers and other technologies within the context of communication. This research has spanned numerous fields and disciplines including engineering, computer science, artificial intelligence (AI), ergonomics, and information science and helped to spur the development of interdisciplinary fields such as computer-supported cooperative work (CSCW) and Human-Computer Interaction (HCI). The term "human-machine communication" has its origins in this scholarship³ and has only recently been increasingly used by communication scholars.

The Role of Communication in Designing Technology

The framing of people's interactions with technology as a type of communication has guided research and development for a wide range of technologies,

including underwater devices (Sheridan & Verplank, 1978), planes (Sarter & Woods, 1995), spaceships (National Aeronautics and Space Administration, 1976), artificial intelligence programs (Weizenbaum, 1966), and computers generally (Licklider, 1960). Within these contexts communication carries a denotation similar to or drawn from cybernetics: it is the transfer of information for the purposes of control (Wiener, 1948). At the same time, communication with and among machines has never been far from a human context in that engineers have used human communication as a guide for designing the points of contact between people and technology, machine interfaces.

Most contemporary technology users take for granted the relative ease with which they control their devices but getting data into early computers and the results out of them was anything but simple or efficient. These hurdles of input and output were framed as problems of communication (e.g., Licklider & Clark, 1962). What Licklider (1960) called “the language problem,” the difference between machine and human languages, was viewed as a barrier to use and adoption. The ontological difference between people, biological beings, and computers, programmed things, also was a formidable communication problem, as NASA states:

Even when people do not speak the same language, they can communicate with one another to some degree (using gestures, facial expressions, etc.) because they share common biological structure and needs, common patterns of thought and behavior, and large overlapping stores of knowledge about the world. Machines and humans do not share these things, and most of the burden of communication falls on the human. (1976, p. 3.4)

As computers became smaller and began moving into everyday spaces, the general difficulty people experienced with adapting to computers continued to be viewed as a communication issue. In their essay regarding computer use as a form of communication, Oberquelle, Kupka, and Maass (1983) argue: “The problems of today’s computer use mainly result from *difficulties in the communication between human and the machine*” (p. 309, emphasis authors).

The solutions devised also were based in communication. The exchange of information was approached as a dialogue (e.g., Meadow, 1970) with human conversation as the prototype. For example, Hayes and Reddy (1979) argue: “We believe that graceful interaction in these and the many other contingencies that can arise in human conversation is essential if interfaces are ever to appear co-operative and helpful ...” (p. 231). Developers also took the relational dynamics of human communication into account, deciding that in interactions, the human, as “master,” would command the machine, as “servant” (e.g., Flanagan, 1976; Mooers, 1959). The development of

language-based interfaces furthered this focus on the use of technology as a communicative process with technology (Suchman, 1990). Some researchers pushed back against using human communication as a prototype, countering that reproducing human interaction within a human-machine context was neither always possible nor desirable (e.g., Nickerson, 1976), but, overall, human communication has served as an influential model for technology design.

Although scholarship across engineering, computer science, and related fields frames certain problems with technology as a lack of or breakdown in communication and turns to human communication for answers, the concept of communication itself and communication theory often are not the focus of this research. What communication is and does is axiomatic, and what communication is all about within these contexts is using technology. The aim of improving people's communication with machines—from the perspective of engineers and computer scientists—is facilitating and promoting technology use.

In an edited volume regarding “man-machine communication,” Meadow (1970) defines the book's key concept as a “two-way conversation that is goal-oriented—aimed toward the accomplishment of a specific objective—and in which both parties contribute a necessary function” (p. 4). That objective, as becomes clear in the remainder of the book, is helping people accomplish tasks. NASA's (1976) definition of “human-machine communication” also is focused on the point of contact between people and technology: “the characteristics of the interface through which a human user instructs or programs a machine, interacts with it during execution, and accepts information from it” (p. 3–79). This general view of the purpose of people's interactions with machines also is integral in Human-Computer Interaction research.⁴ One of HCI's objectives has been and continues to be the development of the human-computer interface, the point of information exchange between human and computer and the processes surrounding it (Baecker, Grudin, Buxton, & Greenberg, 1995). Jacob (2003) explains regarding HCI:

We can view the fundamental task of human-computer interaction as moving information between the brain of the user and the computer. Progress in this area attempts to increase the useful bandwidth across that interface by seeking faster, more natural, and more convenient communication means. (p. 145)

Theorizing Communication in Technology Use

From the standpoint of communication theory, this research regarding people's interactions with machines proceeds from what Carey (1989) called a transmission view of communication. Communication is about the exchange of information toward some desired effect. The machine is theorized as having a degree of agency in that it performs a distinct role during an interaction

(Fischer, 1990; Meadow, 1970) and draws from its own resources in processing and responding to messages (Riley, 1989). But, its ontology is very much tied to its status as a technology and the predominant cultural conceptualization of technology as tools.⁵ Human-machine communication within this context is a process of interacting with technology, a tool, as to leverage it for some purpose.

What has been largely absent from this body of work in relation to communication theory is the social aspect of communication, including how technologies figure into people's social worlds and the implications for society (Zhao, 2006). The subjectivity of the technology is limited: its place within a user's social world is extremely restricted, if acknowledged at all. Outside its use, the technology has no bearing on how people understand their world and themselves. Communication with technology is an "isolated" process without implications beyond what transpires within the interaction.

There are notable exceptions such as Suchman's (1987) foundational work in human-machine communication that approaches people's interactions with technology as "situated actions," collaboratively unfolding between human and machine: "Communication in this sense is not a symbolic process that happens to go on in real-world settings, but a real-world activity in which we make use of language to delineate the collective relevance of our shared environment" (p. 180). Suchman (2009) also draws on scholarship from feminist theory and Science and Technology Studies (STS) to interrogate the social and cultural dimensions of human-machine communication.⁶ More recently HCI scholars also have taken up critical and cultural questions; although, reconciling critical and cultural work with the practical aims of HCI research to inform design and facilitate use has proved challenging (Rogers, 2012).

Overall, communication has played an integral role in the development of technology and its use. The technologies that are now inspiring communication scholars to pay attention to human-machine communication are the result of decades of research guided by the idea of communication outside the discipline. Researchers in HCI have made important strides in the transmission of information from human to machine and facilitating technology use. However, with a focus on communication within this context, other aspects of communication with technology have received less attention, including social aspects and cultural implications.

Medium as Message and Messenger

While scholars outside the discipline initially propelled human-machine communication research, communication scholars have not been entirely out of

the picture. Communication scholars have directly contributed to HCI and related areas within the past few decades. There also are some theoretical traditions within the discipline that have focused their research directly on the medium and have potential for furthering the study of people's interactions with machines and addressing existing theoretical oversights.

Communication Engineering vs Human Communication

Some of the fields whose literature was assimilated by communication scholars were the same fields focused on technology research and development, including cybernetics and information theory (Craig, 1999; Rogers, 1997). Cybernetics defined communication as a process of information transfer among people, animals, and machines (Wiener, 1948): Who or what were involved in the interaction were not the defining element of communication in cybernetics—the process was.⁷ Similarly Shannon and Weaver's (1949) theory, considered to be communication's foundational model, evolved from a theoretical context similar to cybernetics.⁸ As originally explicated by Shannon (1948), the model included sending messages between people and machines. And so, some of the literature that would inform communication's early theories conceptualized communication as a process between humans as well as humans and machines.

Scholars who were helping to shape communication, however, decided that the discipline would focus on human interaction.⁹ Existing theories were retrofitted to a human-only context, including Shannon and Weaver's (1949) model (Rogers, 1997). As an engineer, Shannon originally wrote *A Mathematical Theory of Communication* (Shannon, 1948) as a theory of communication engineering. Shannon's focus was maximizing the transmission of signals from one point (a human or machine) to another point (a human or machine). As Rogers (1997) explains, Shannon's original model had nothing to do with meaning-making. Shannon's theory often is attributed to both Shannon and Weaver because it was later republished in a book with a similar title and an introduction by Weaver. According to Rogers (1997), this introduction (Weaver, 1949) adds to and adapts Shannon's theory from an engineering to social science context, recasting it as a means to study the effects of messages. The specific elements of the model that within engineering could be occupied by a human or a machine came to be associated with humans-only (Rogers, 1997). The distinct roles for humans and machines became codified and were reinforced in subsequent models (e.g., Barnlund, 1970; Lasswell, 1972): humans are communicators (senders and receivers) and technology is the medium, or channel, through which people exchange messages. This has been the dominant paradigm for communication research.